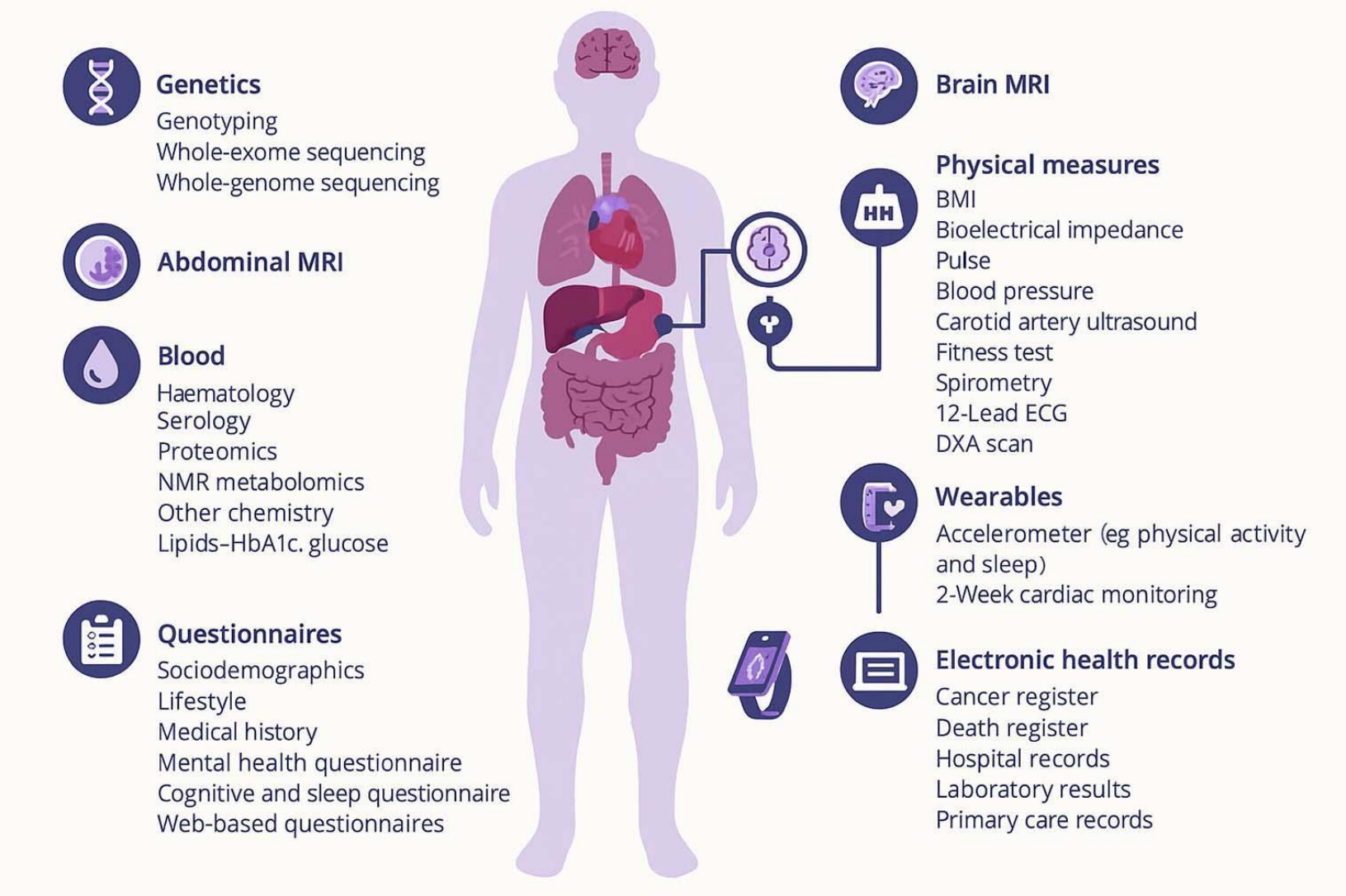
Humans are becoming a major model system for quantitative biology



https://www.ukbiobank.ac.uk/

Single-cell technologies are revealing the cellular diversity of human tissues

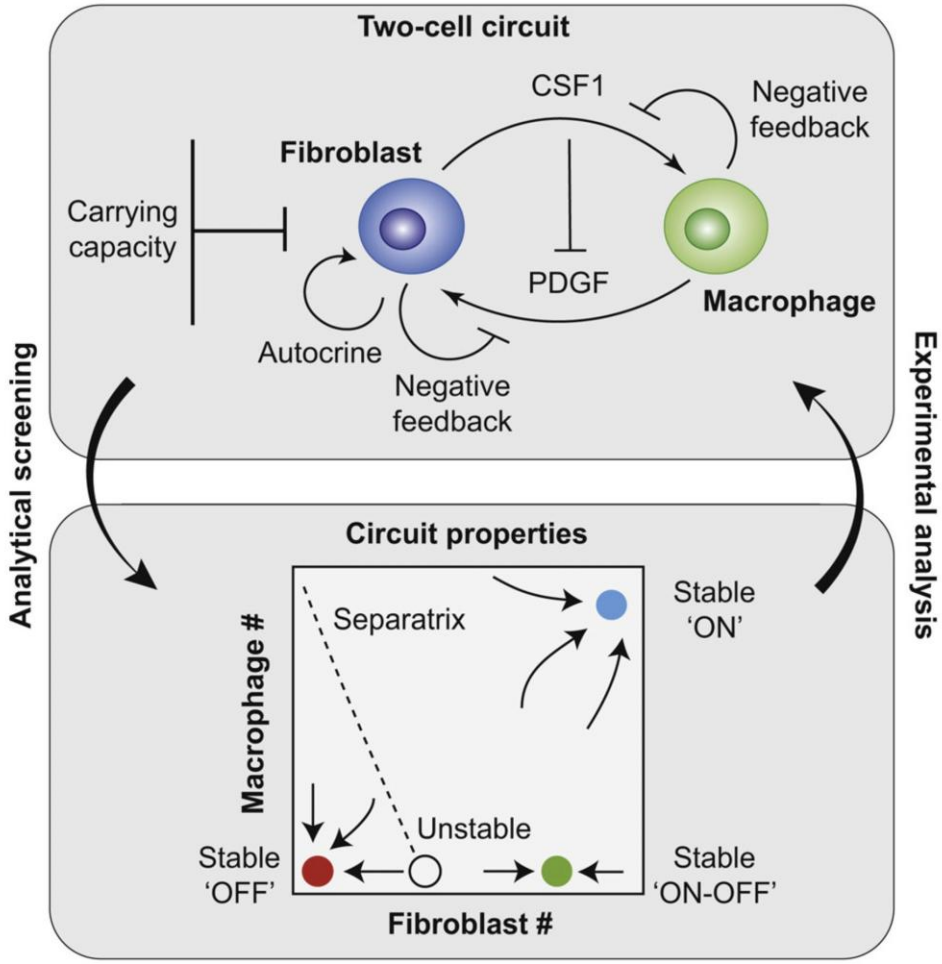


- AT0
- AT1
- AT2
- AT2 proliferating
- Adventitial fibroblasts
- Alveolar Mph CCL3+
- Alveolar Mph MT-positive
- Alveolar Mph proliferating
- Alveolar fibroblasts
- Alveolar macrophages
- B cells
- Basal resting
- CD4 T cells
- CD8 T cells
- Classical monocytes
- Club (nasal)
- Club (non-nasal)
- DC1
- DC2
- Deuterosomal
- EC aerocyte capillary
- EC arterial
- EC general capillary
- EC venous pulmonary
- EC venous systemic
- Goblet (bronchial)
- Goblet (nasal)
- Goblet (subsegmental)
- Hematopoietic stem cells
- Hillock-like
- Interstitial Mph perivascular
- Ionocyte
- Lymphatic EC differentiating
- Lymphatic EC mature
- Lymphatic EC proliferating
- Mast cells
- Mesothelium
- Migratory DCs
- Monocyte-derived Mph
- Multiciliated (nasal)
- Multiciliated (non-nasal)
- Myofibroblasts
- NK cells
- Neuroendocrine
- Non-classical monocytes
- Peribronchial fibroblasts
- Pericytes
- Plasma cells
- Plasmacytoid DCs
- SM activated stress response
- SMG duct
- SMG mucous
- SMG serous (bronchial)
- SMG serous (nasal)
- Smooth muscle
- Smooth muscle FAM83D+
- Subpleural fibroblasts
- Suprabasal
- T cells proliferating
- Tuft
- pre-TB secretory
- NA

Human Lung Cell Atlas, single-cell UMAP visualization

Knowing which cells are present does not tell us how a tissue will evolve

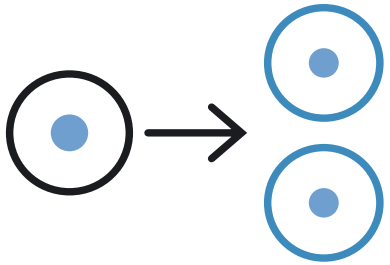




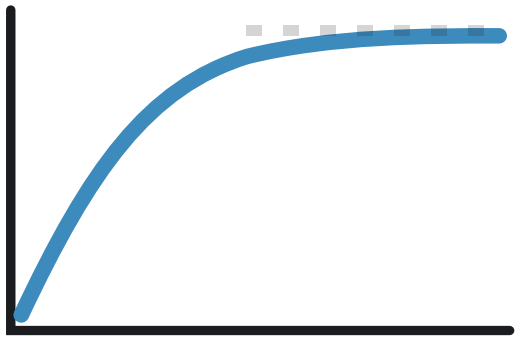
Human Lung Cell Atlas, single-cell UMAP visualization

Three laws of physiology constrain the dynamics of tissues

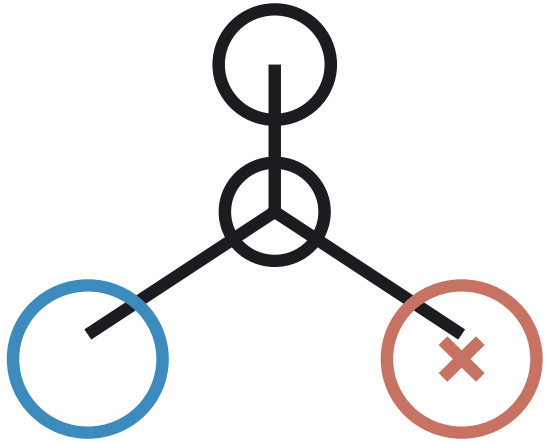
1. All cells come from cells



2. Biological processes saturate

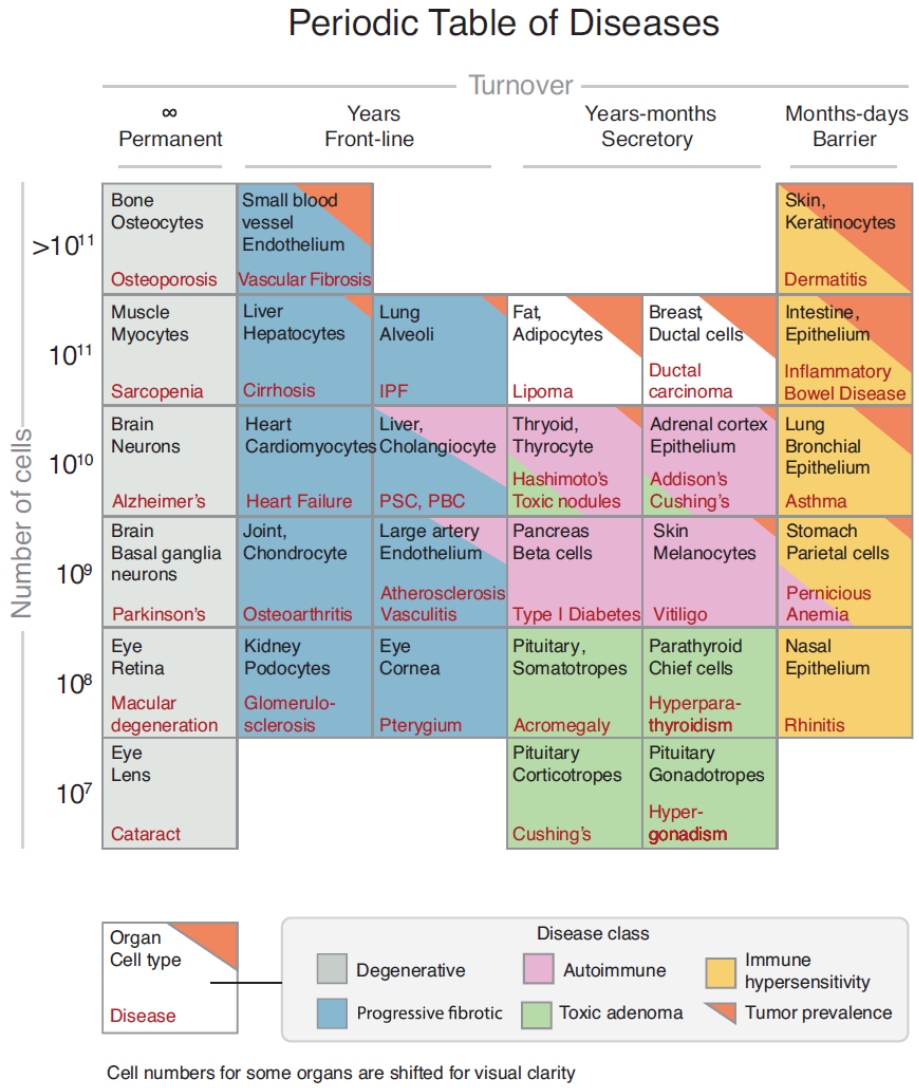


3. Cells mutate



Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

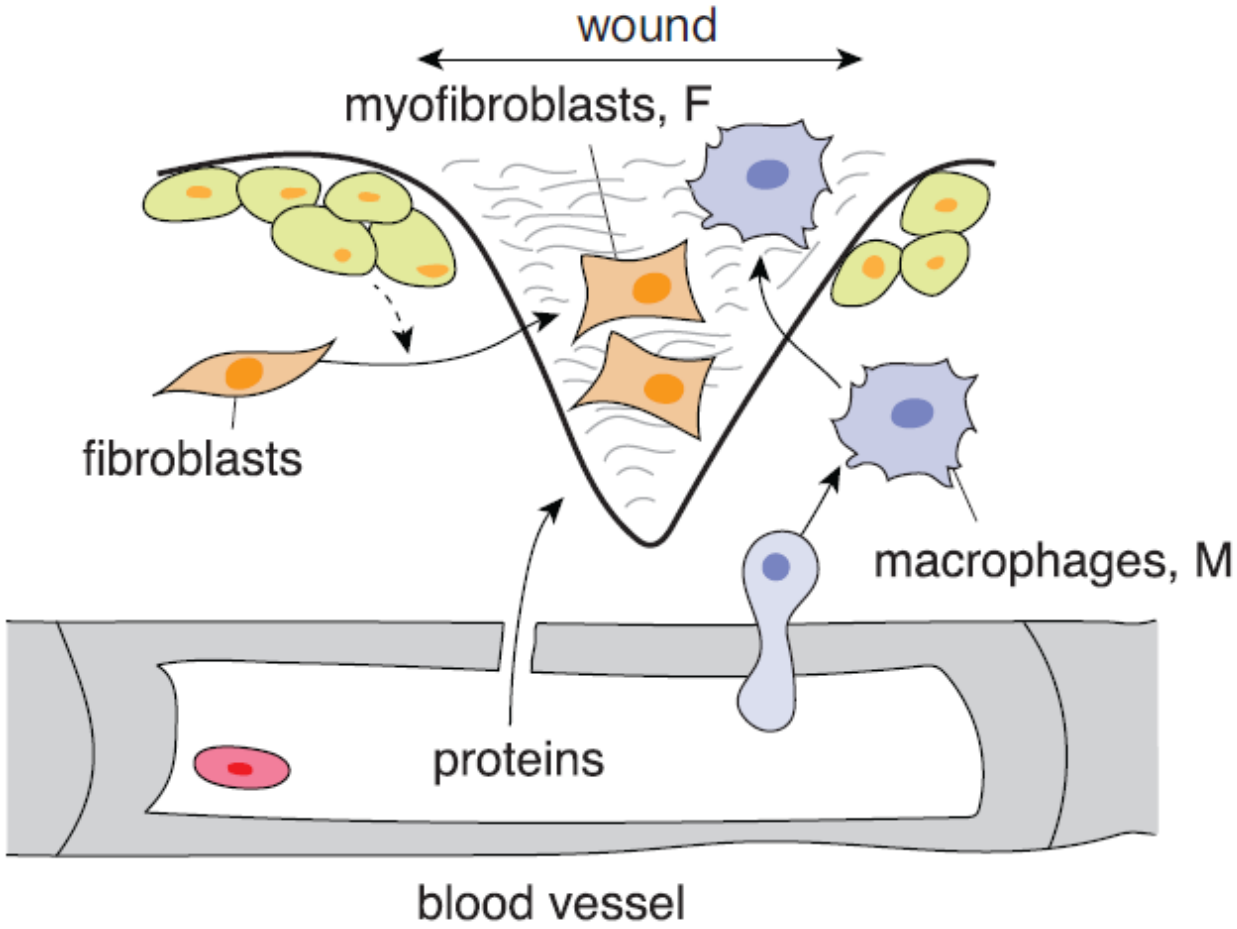
Simple physiological laws can organize the landscape of human diseases



Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

There are 3 specific observations regarding fibrosis and 2 central questions

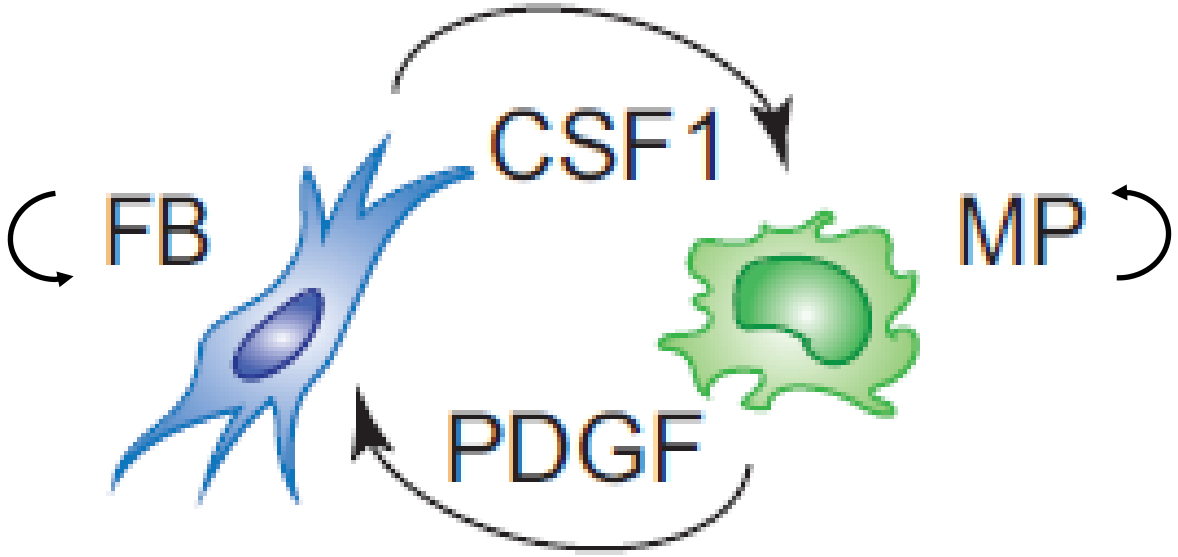
Inflammation involves redness, swelling, heat and pain



Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

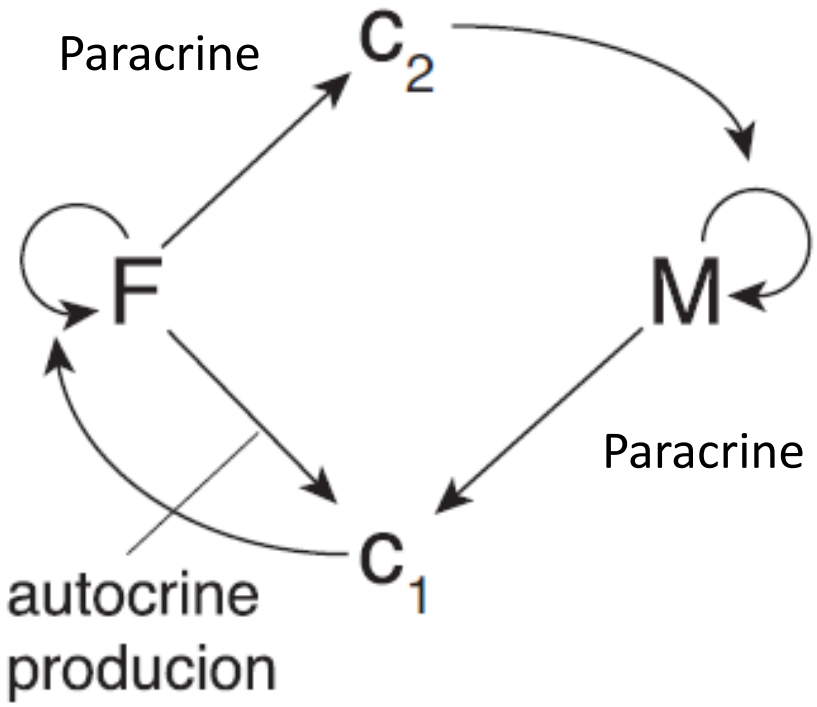
Fibrosis is very complex but our goal is to construct the essential conceptual picture and add back the details later

Myofibroblasts and macrophages secrete growth factors to support each other



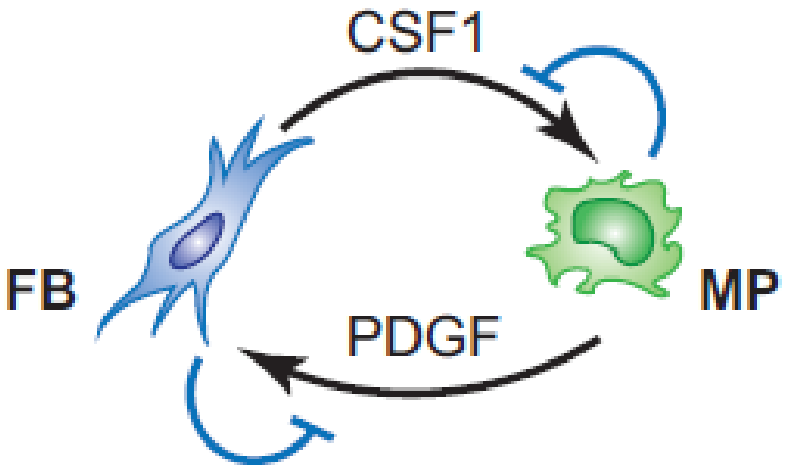
Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

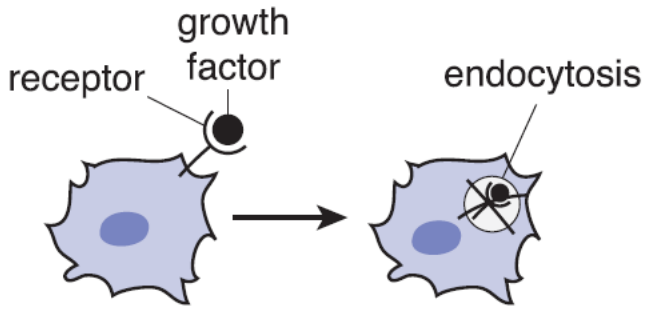
The paracrine and autocrine loop between myofibroblasts (F) and macrophages (M) form a cell circuit



Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

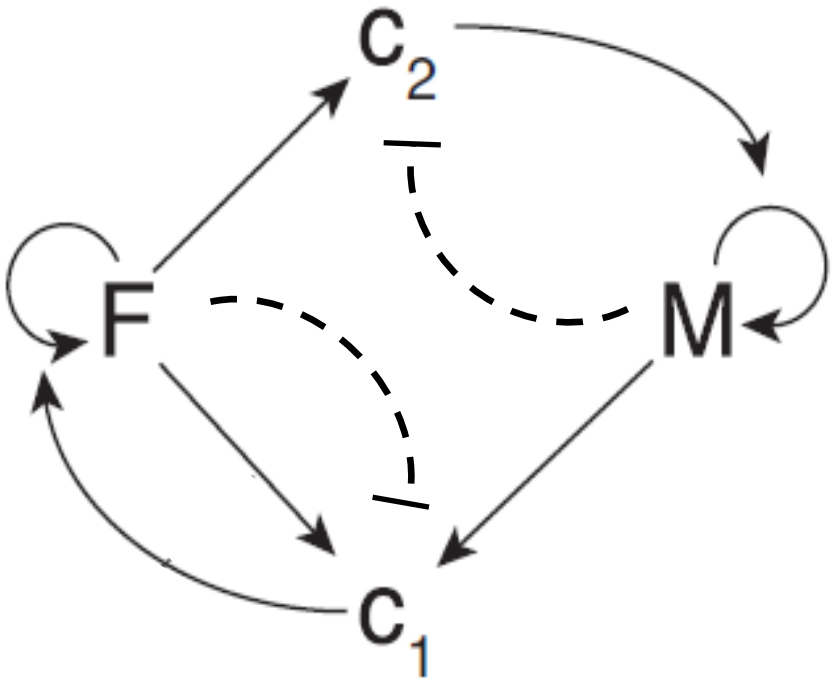
Growth factors are removed via endocytosis by their targets





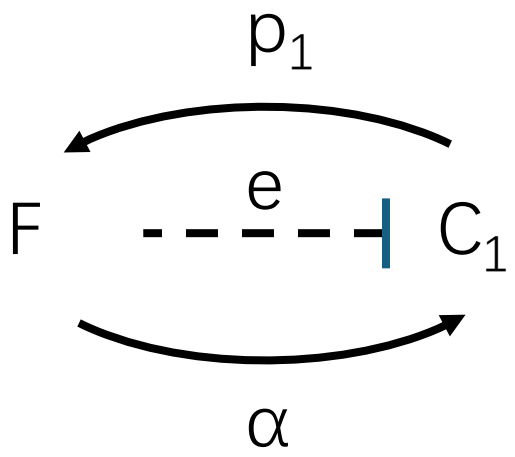
Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

We can write a mathematical model for the fibrosis circuit by keeping track of M, F, c1 and c2



Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

We can focus on F for now and add M back later



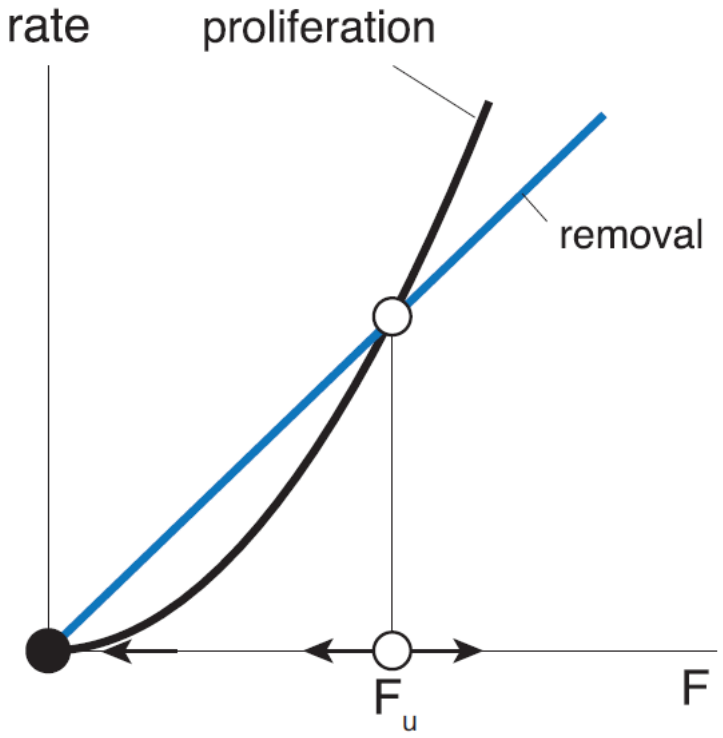
$$\frac{dC_1}{dt} = aF - eFC_1 - \gamma C_1$$

Autocrine Endocytosis Degradation

$$\frac{dF}{dt} = p_1 F C_1 - d_1 F$$

$$C_1 = \frac{aF}{eF + \gamma}$$

We can use the rate plots of F proliferation and removal to understand dynamics of F by itself



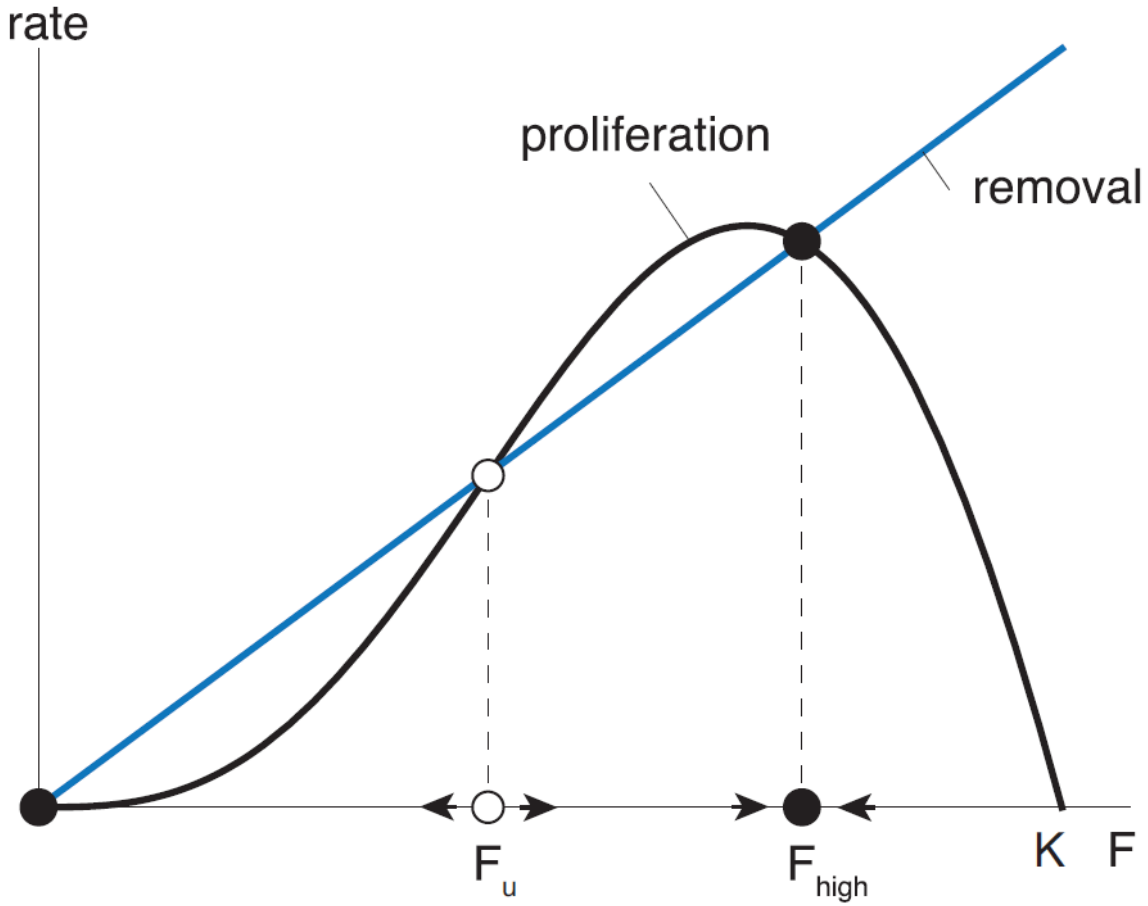
$$\frac{dF}{dt} = p_1 F C_1 - d_1 F$$

$$C_1 = \frac{aF}{eF + \gamma}$$

$$\frac{dF}{dt} = p_1 F \left(\frac{aF}{eF + \gamma} \right) - d_1 F$$

Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

But experimental observations establish a carrying capacity for F

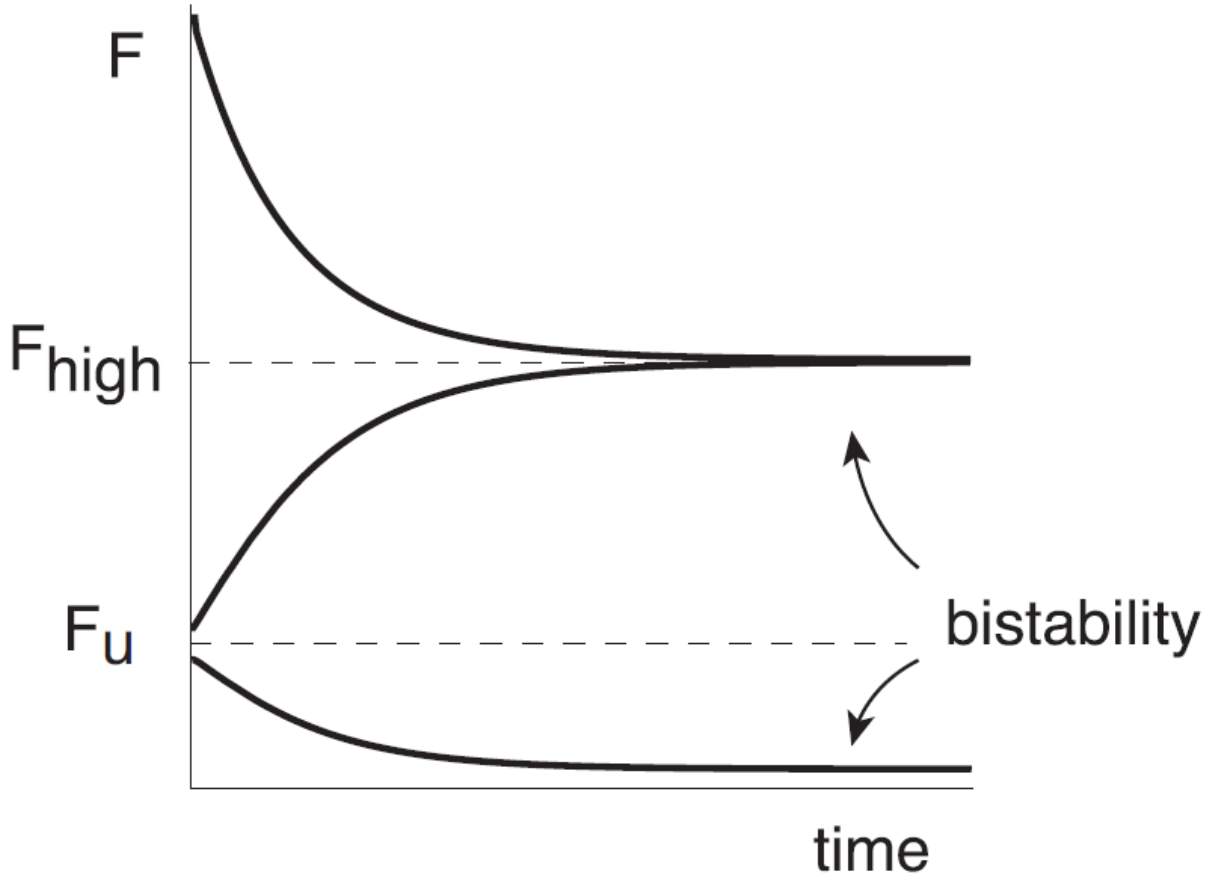


$$\frac{dF}{dt} = p_1 C_1 \left(1 - \frac{F}{K}\right) F - d_1 F$$

Max Capacity K

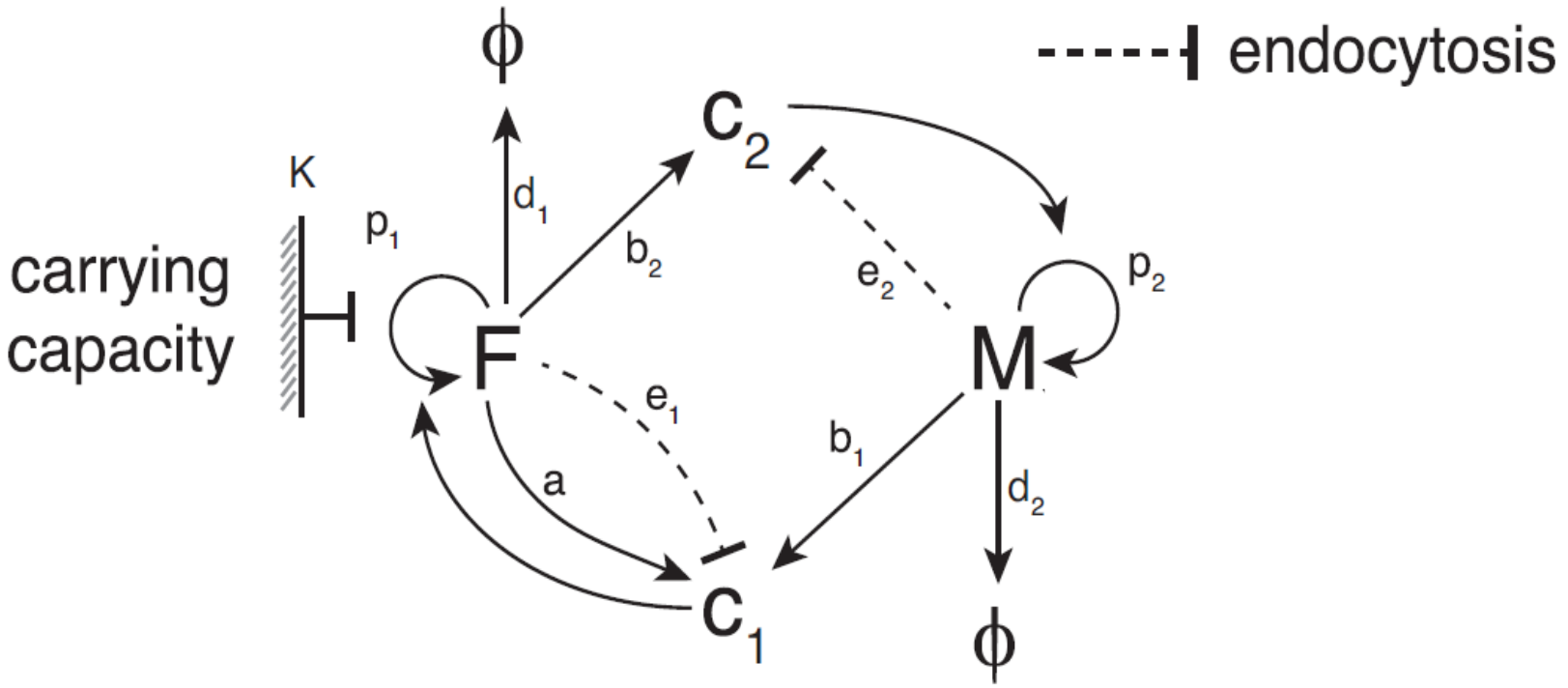
Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

Bistability can be observed experimentally by seeding 96 well plates with varying initial number of F cells



Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

Now we are ready to add M and c2 to the system to understand the full picture



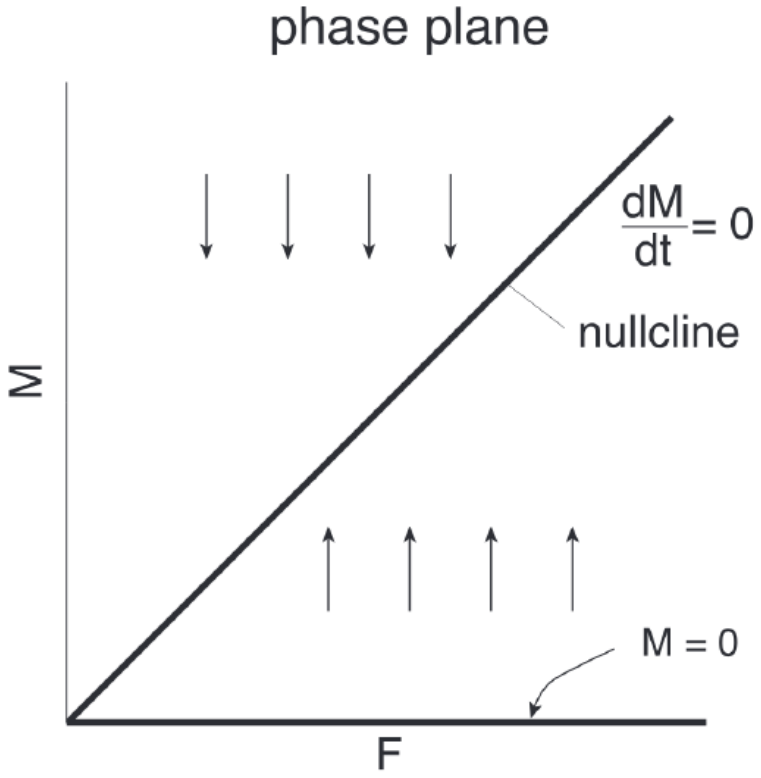
Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

The c2 equation has very similar form, while the M nullcline gives a linear solution

$$\frac{dC_2}{dt} = b_2F - e_2MC_2 - \gamma_2C_2$$

$$c_2 = \frac{b_2F}{e_2M + \gamma_2}$$

$$\frac{dM}{dt} = p_2MC_2 - d_2M$$



Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

The c1 equation has an additional term, which give rise to a U-shaped F nullcline

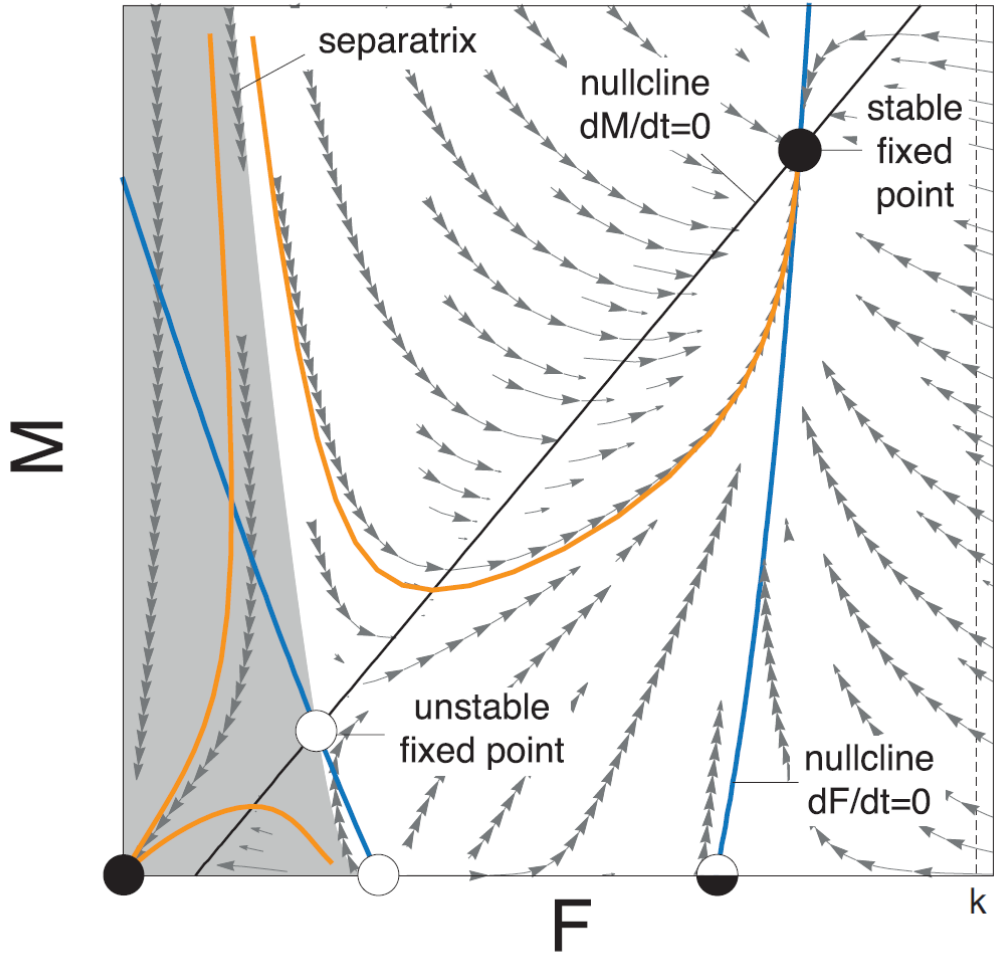
$$\frac{dC_1}{dt} = aF + b_1 M - e_1 F C_1 - \gamma C_1$$

$$c_1 = \frac{aF + b_1 M}{e_1 F + \gamma_1}$$

$$\frac{dF}{dt} = p_1 C_1 \left(1 - \frac{F}{K}\right) F - d_1 F$$

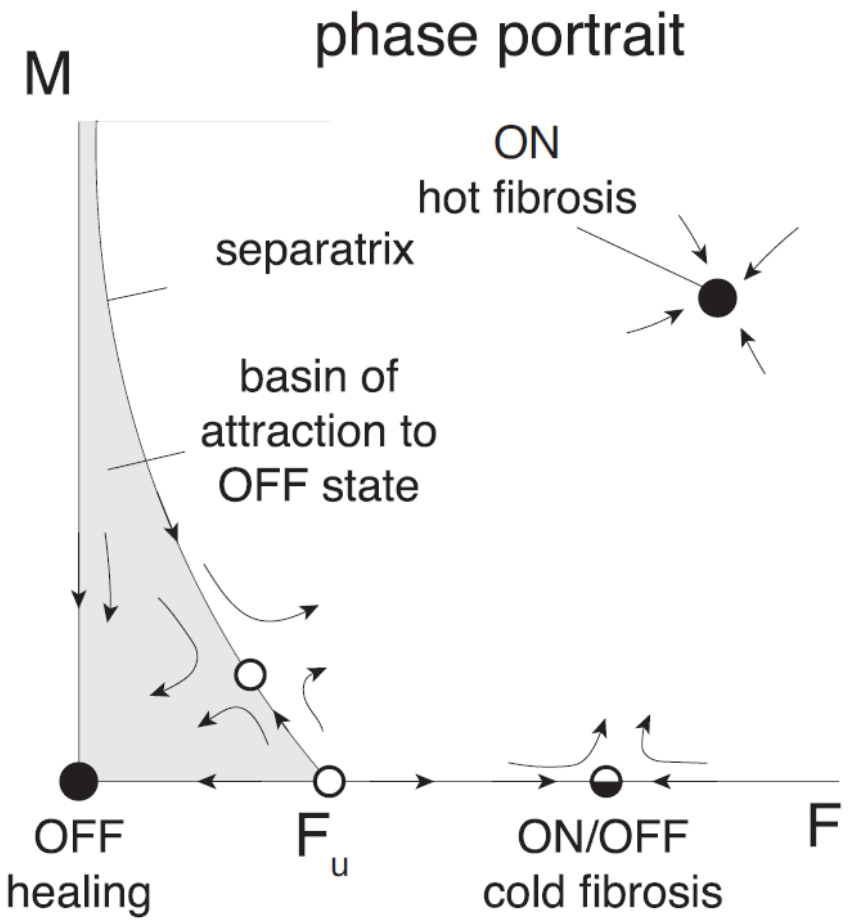
Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

The M and F nullclines help us plot the fixed points and vector field on M-F phase plane



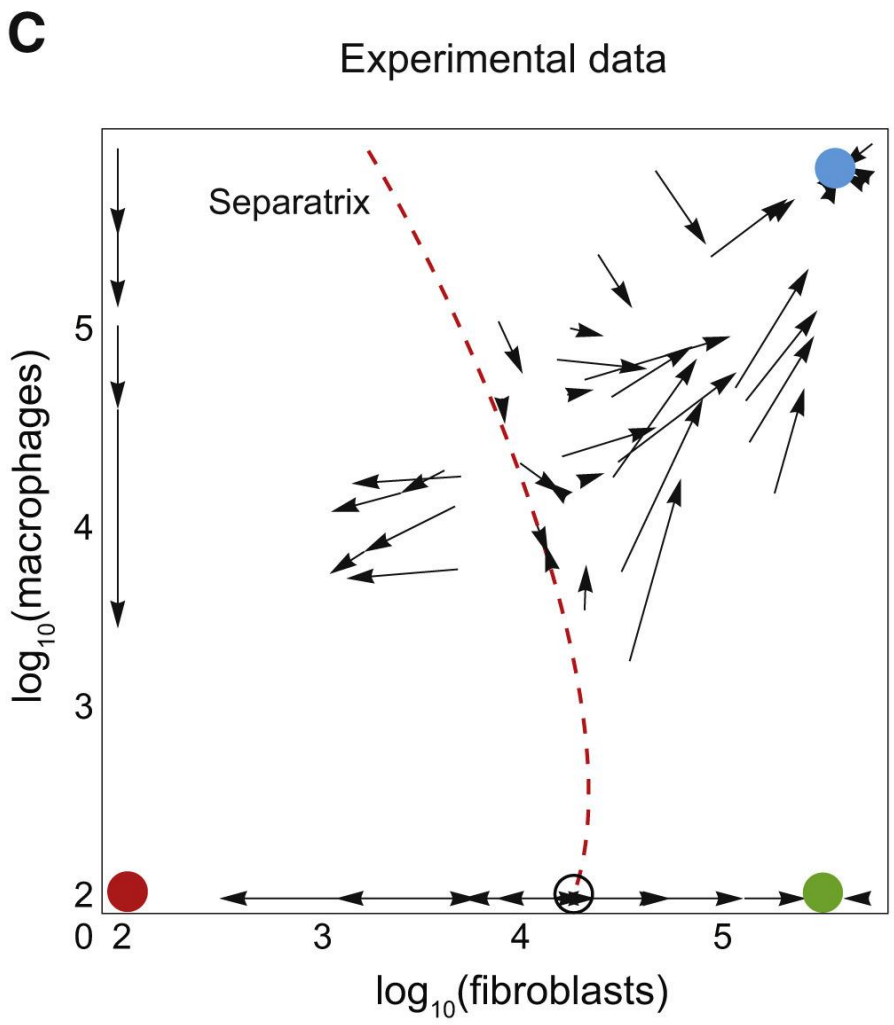
Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

The phase portrait on M-F plane gives us the qualitative picture of the fibrosis circuit



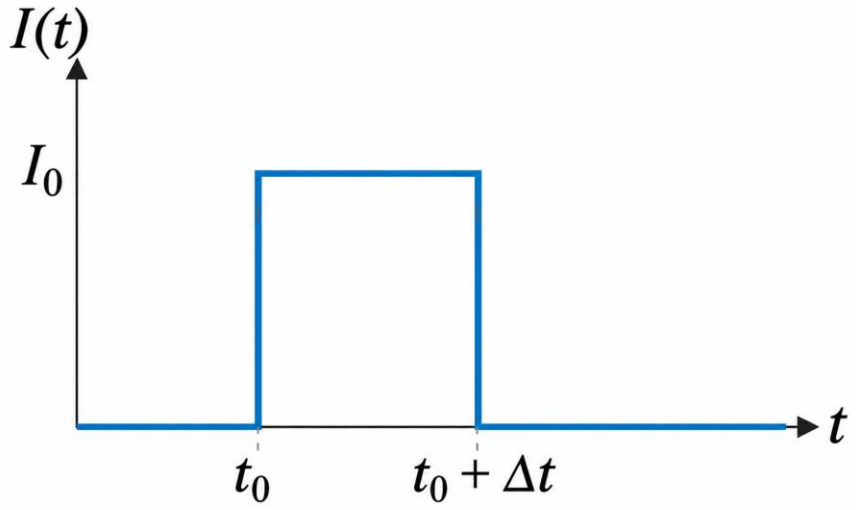
Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

The fibrosis circuit is figured out experimentally by Xu Zhou et al, using cell cultures from varying initial densities



Zhou X, Franklin R, Adler M, …, Alon U and Medzhitov R (2018) Cell

Injury and inflammation can be modelled by a transient influx of macrophases



$$\frac{dC_2}{dt} = b_2 F - e_2 M C_2 - \gamma_2 C_2$$

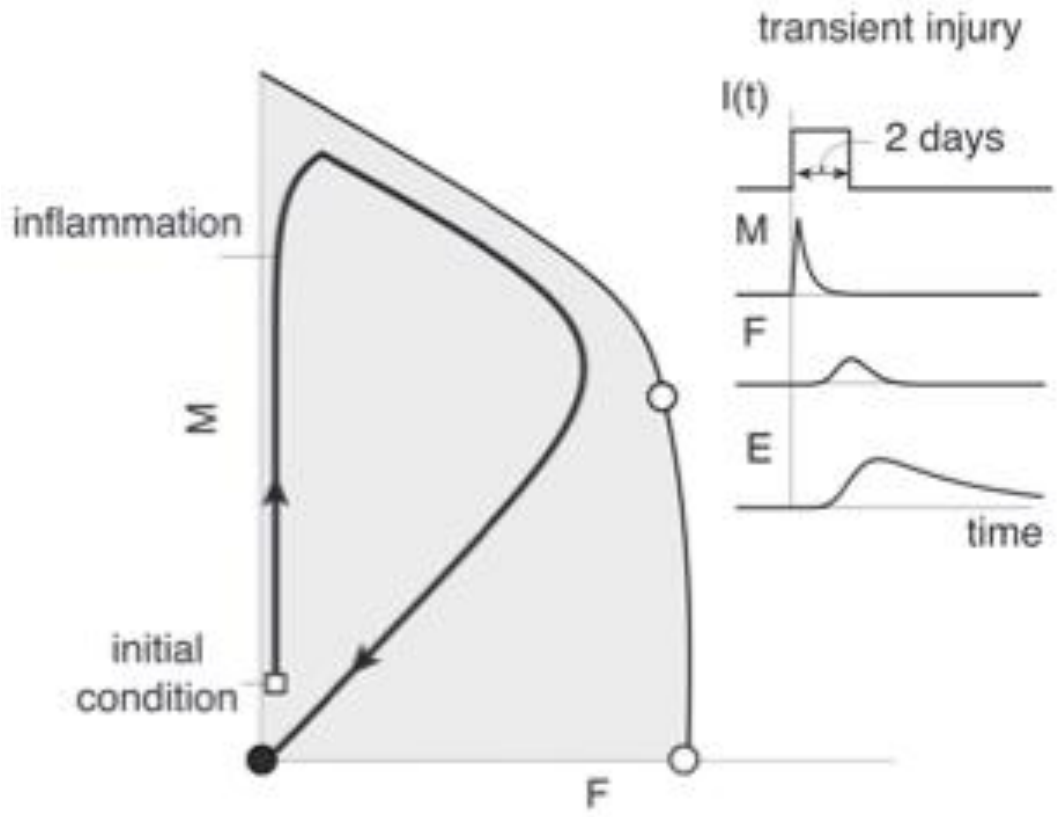
$$\frac{dM}{dt} = p_2 M C_2 - d_2 M + I(t)$$

$$\frac{dC_1}{dt} = aF + b_1 M - e_1 F C_1 - \gamma C_1$$

$$\frac{dF}{dt} = p_1 C_1 \left(1 - \frac{F}{K}\right) F - d_1 F$$

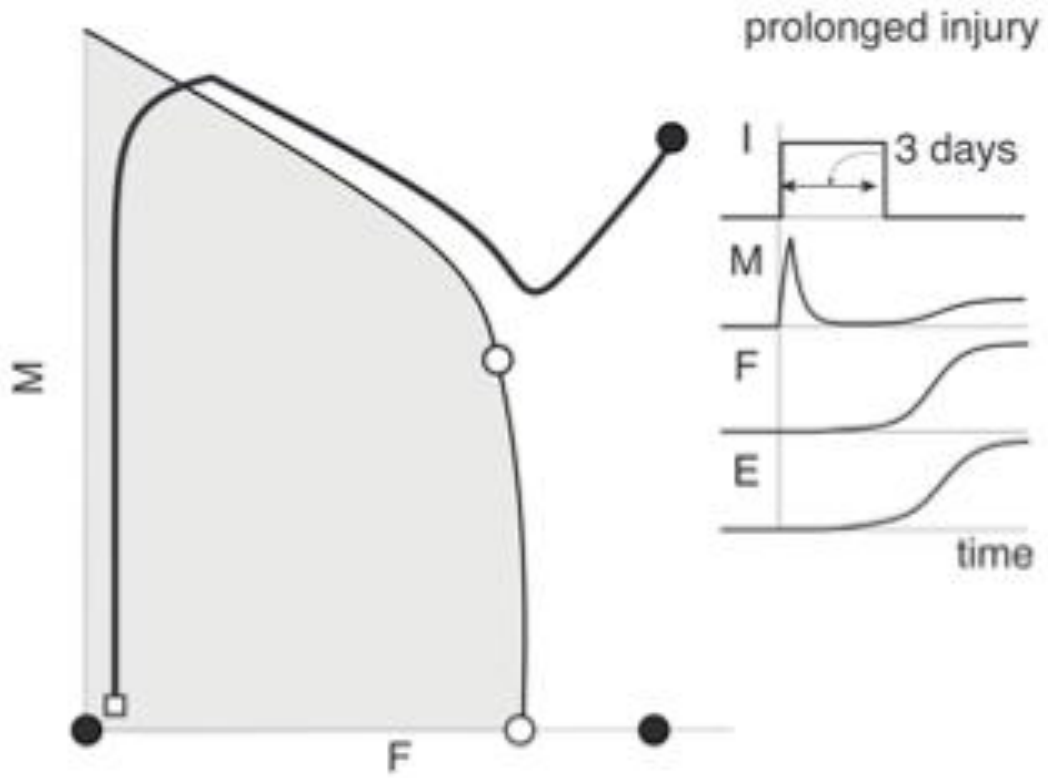
Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

A transient injury causes influx of macrophages but the system remain in the basin of attraction of the OFF state



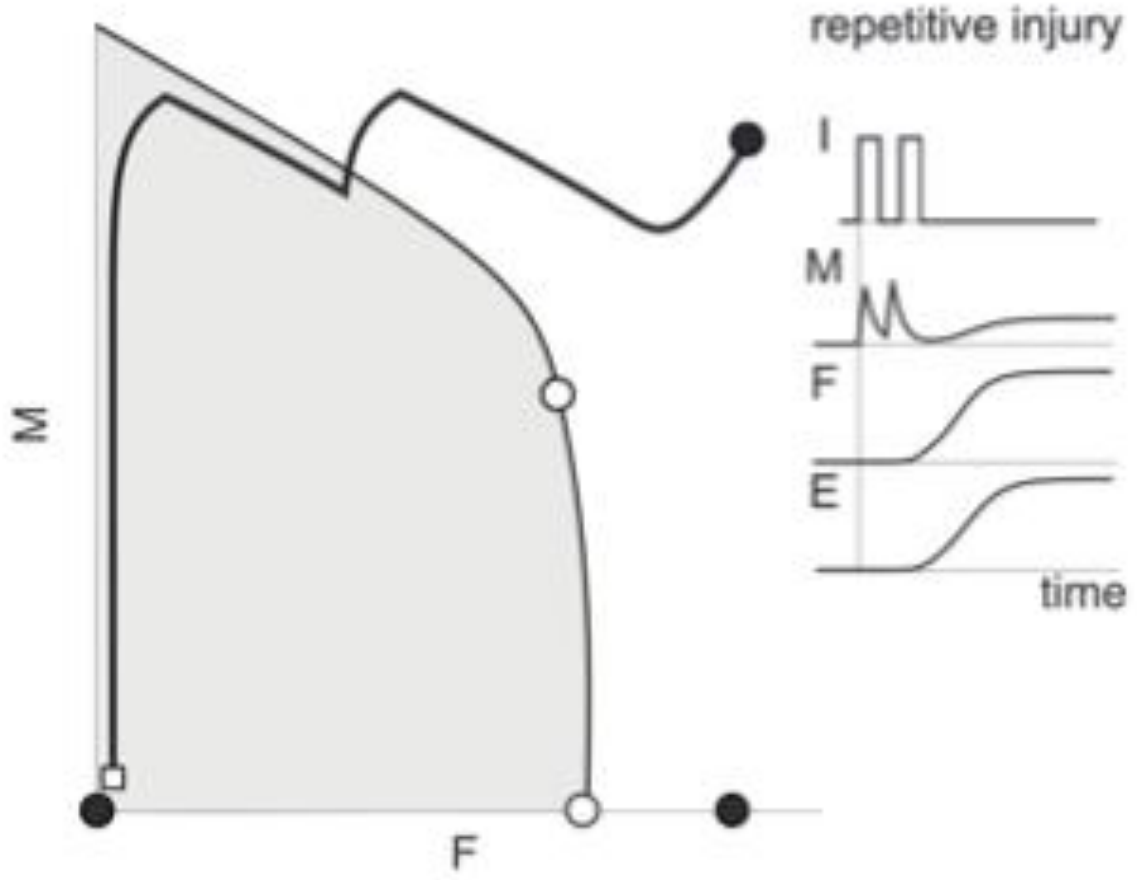
Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

A prolonged injury crosses the separatrix into the basin of attraction of the ON state



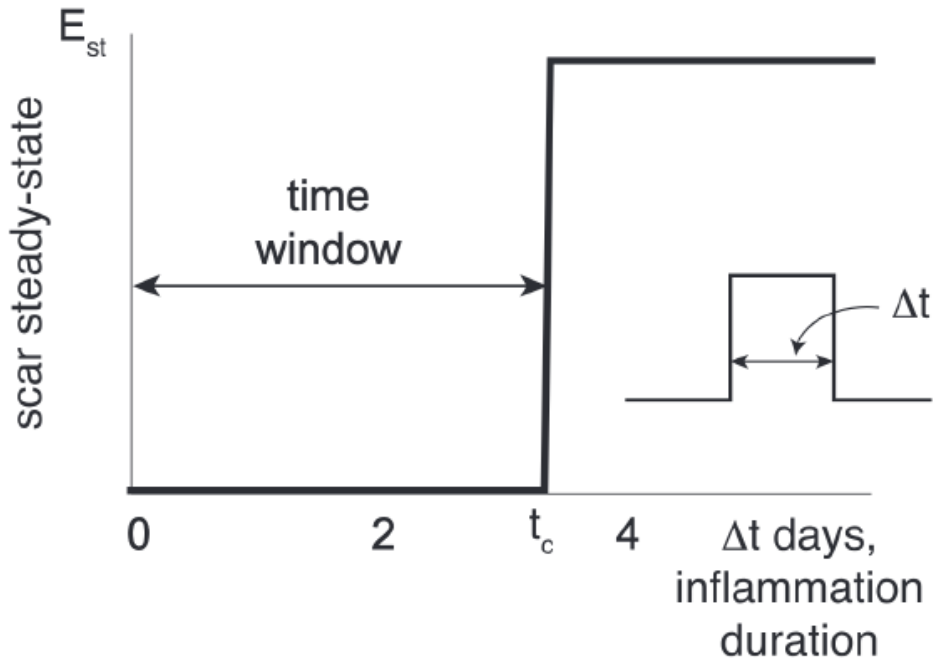
Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

So do reocurring injuries



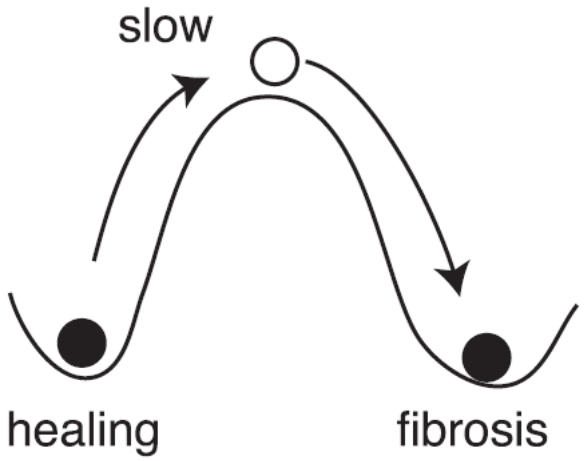
Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

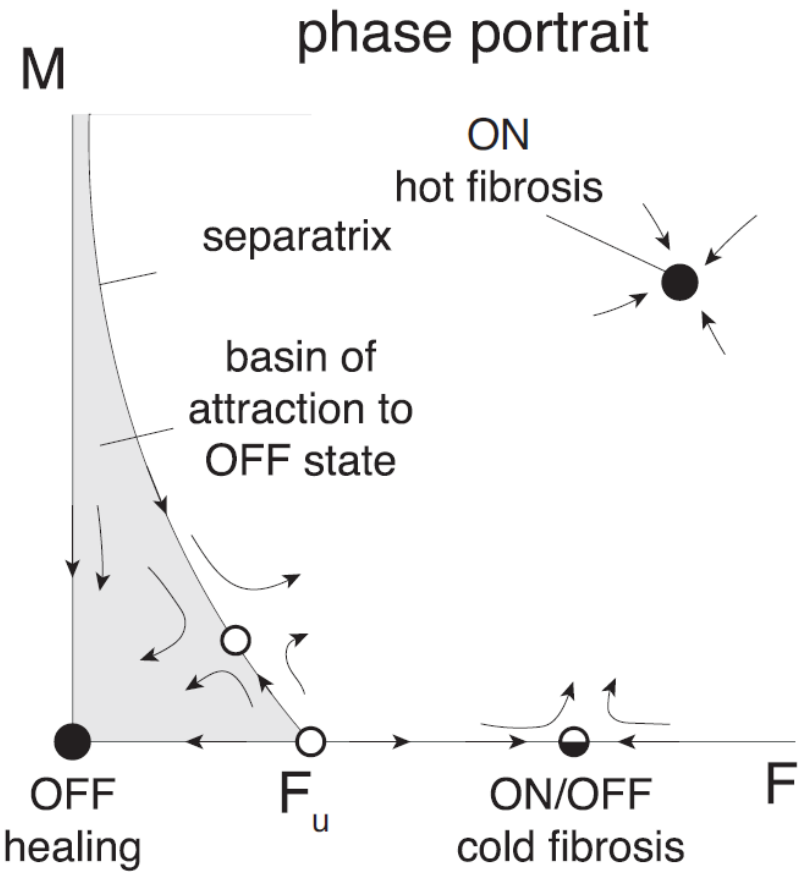
The golden window for preventing fibrosis is due to bistability



Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

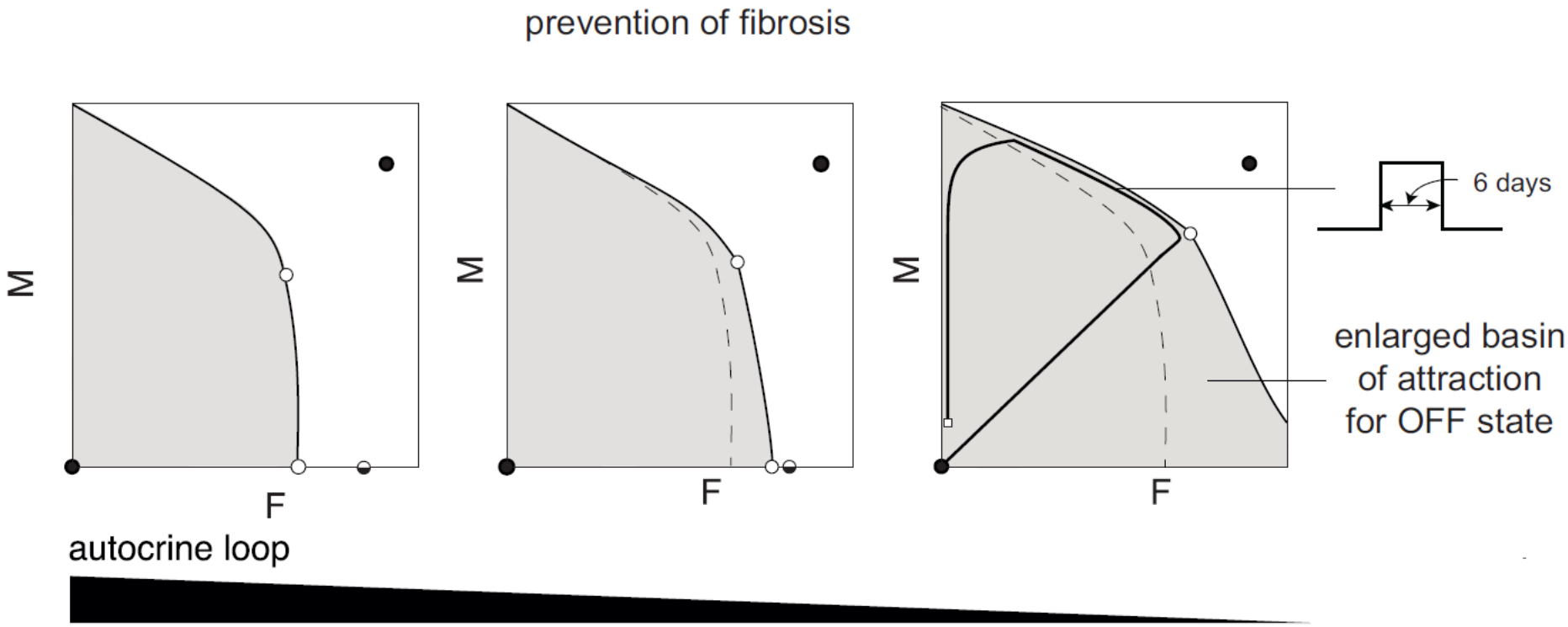
The long timescale for scar maturation and healing is due to the slowdown near an unstable fixed point.





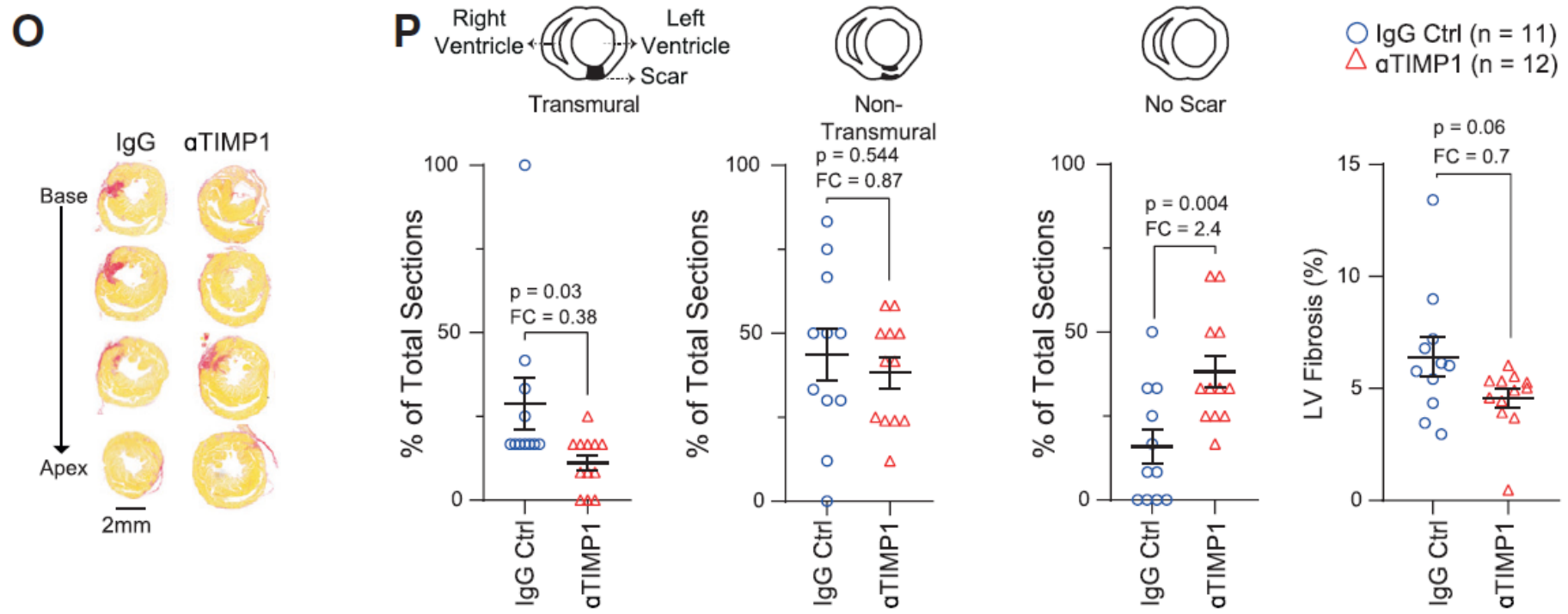
Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

Strategies that reverse fibrosis involve fold bifurcation of F as well as depletion of macrophages



Alon, U. (2023). Systems Medicine: Physiological Circuits and the Dynamics of Disease (1st ed.).

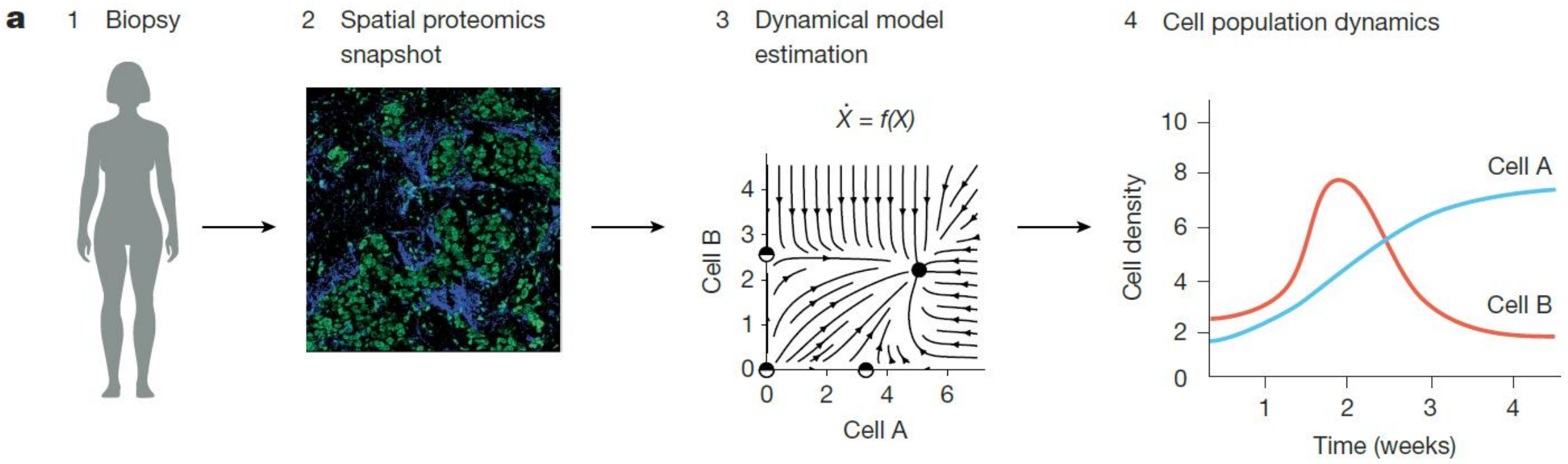
A cell-circuit model guided the discovery of a treatment that blocks myofibroblast autocrine signaling and reduces heart fibrosis in mice



Miyara et al. (2025) Cell Systems

In biopsy for cancer, similar methods can be used to infer the dynamics of cells

One-shot tissue dynamics reconstruction (OSDR)

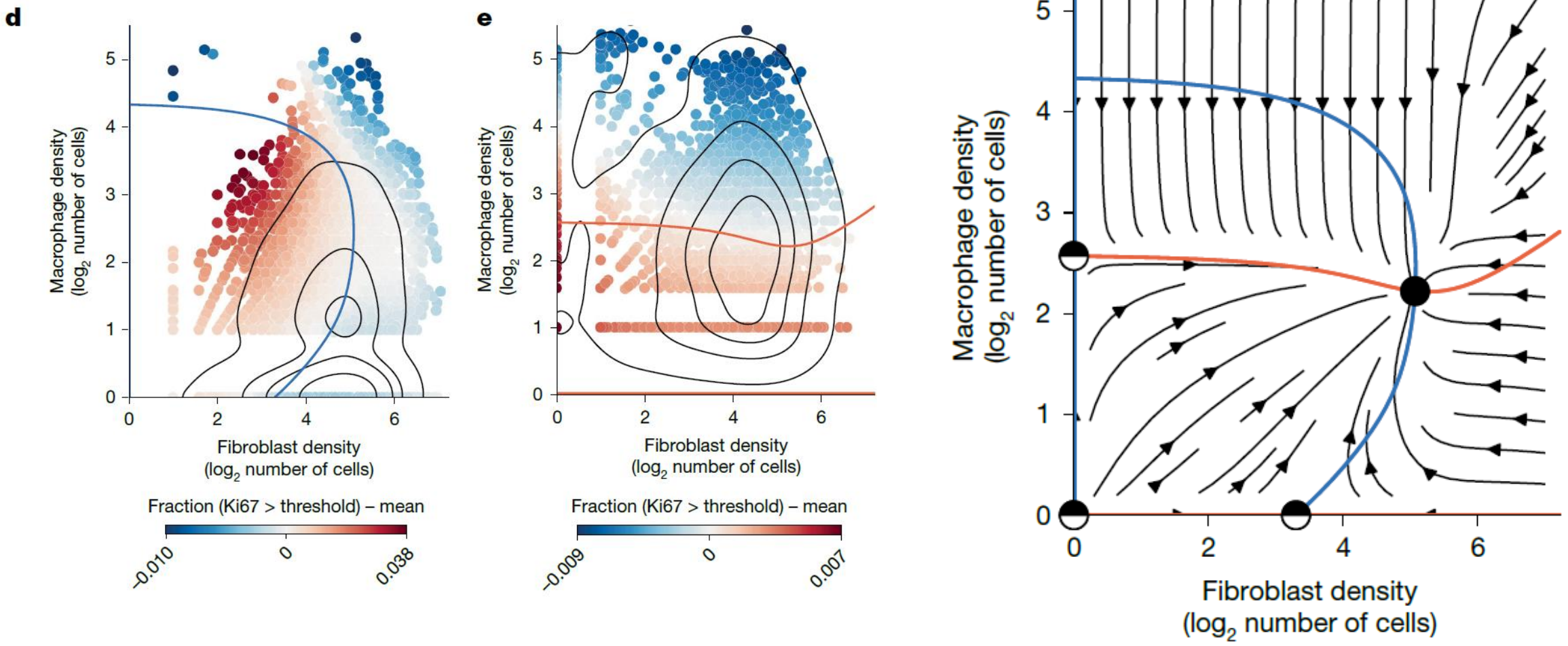


Somer et al. Nature, (2026).

The dynamics of the macrophages and fibroblasts obtained using this method are similar to those in the previous study

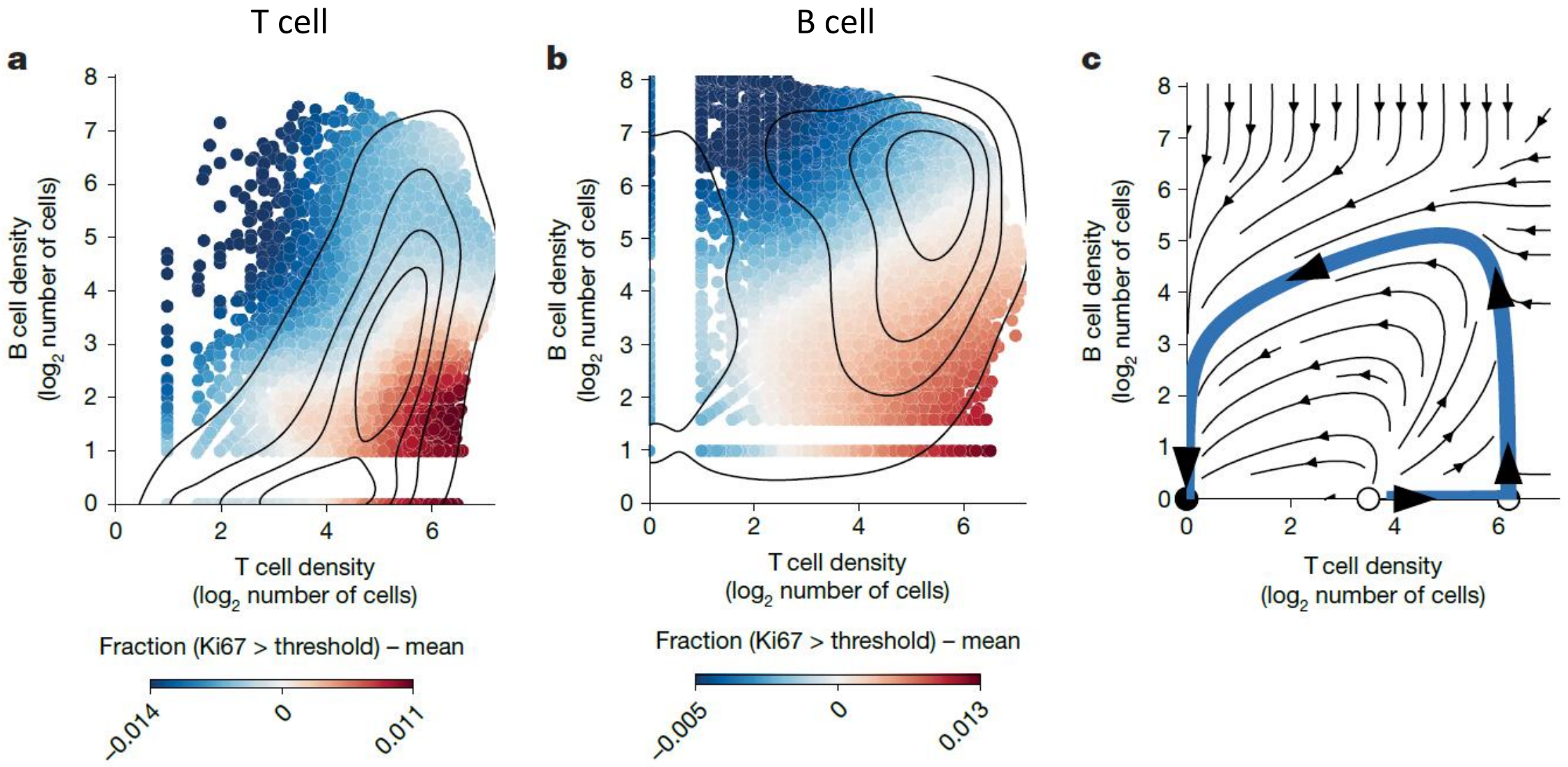
Fibroblasts

Macrophages



Somer et al. Nature, (2026).

Sommer et al. used this method to discover a pulse-generating excitable circuit between T cell and B cell.



Somer et al. Nature, (2026).

Cell-population dynamics inferred from responders and nonresponders predict tumor clearance and persistence, respectively

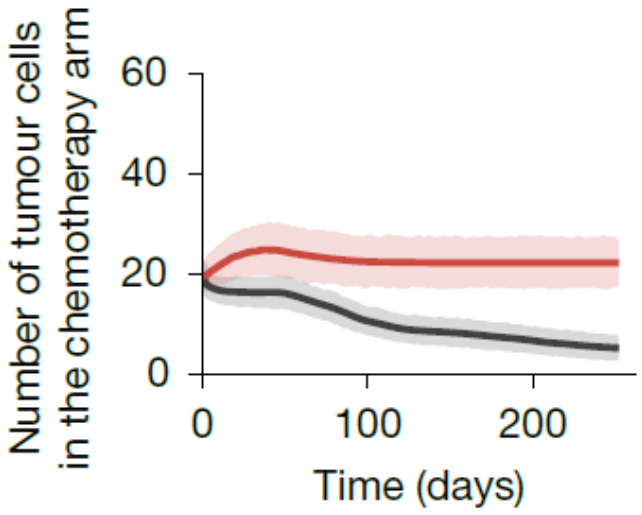
Samples from the third weeks

$$\mathbf{X} = (C, F, M, T, B, E)$$

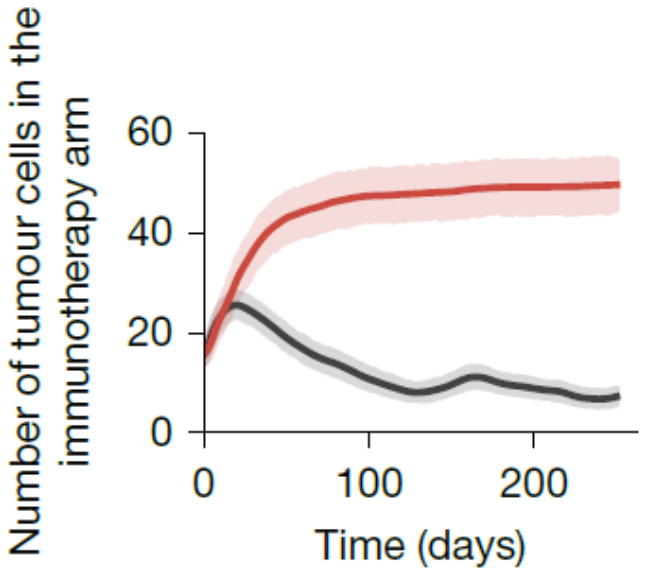
$$\frac{d\mathbf{X}}{dt} = \mathbf{f}_{\mathrm{NR}}(\mathbf{X})$$

$$\frac{d\mathbf{X}}{dt} = \mathbf{f}_{\mathrm{R}}(\mathbf{X})$$

Chemotherapy



Chemotherapy and immunotherapy



Response dynamics

No-response dynamics

Somer et al. Nature, (2026).